EXP-5

Lexicographically Smallest String Using Raptor

Aim

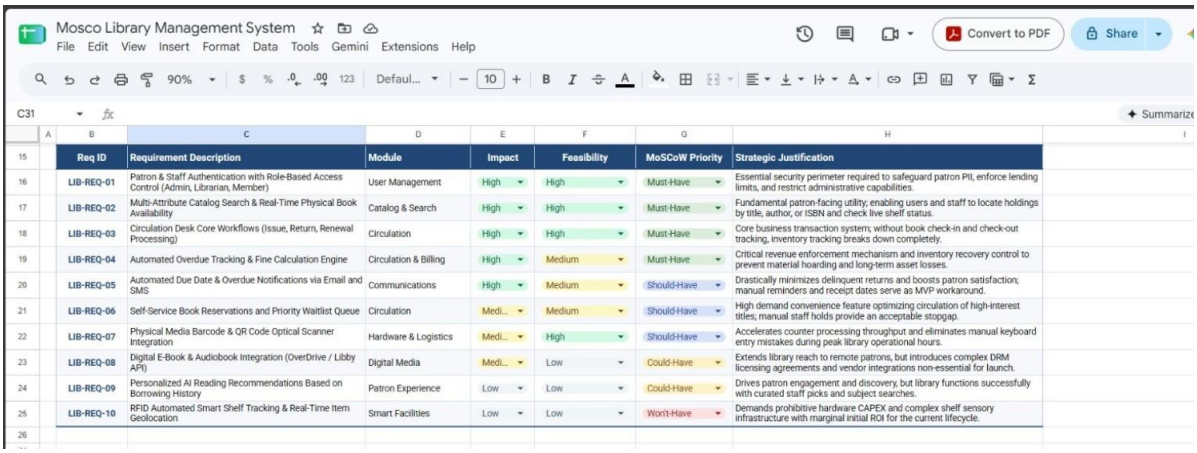
To design a flowchart using Raptor to obtain the lexicographically smallest string by swapping two characters at most once.

Procedure

1. Open Raptor and create a new flowchart.
2. Design the flowchart components: Input String, Identify Characters for Swapping, Perform Character Replacement, and Display Final String.
3. Accept a lowercase string from the user.
4. Find two characters whose swapping gives the smallest lexicographical order.
5. Replace all occurrences of the first character with the second character and vice versa.
6. Generate and display the final modified string.
7. Execute the flowchart and validate the output.

Output

Output – Raptor Flowchart



Req ID	Requirement Description	Module	Impact	Feasibility	MoSCoW Priority	Strategic Justification
LIB-REQ-01	Patron & Staff Authentication with Role-Based Access Control (Admin, Librarian, Member)	User Management	High	High	Must-Have	Essential security perimeter required to safeguard patron PII, enforce lending limits, and restrict administrative capabilities.
LIB-REQ-02	Multi-Attribute Catalog Search & Real-Time Physical Book Availability	Catalog & Search	High	High	Must-Have	Fundamental patron-facing utility; enabling users and staff to locate holdings by title, author, or ISBN and check live shelf status.
LIB-REQ-03	Circulation Desk Core Workflows (Issue, Return, Renewal Processing)	Circulation	High	High	Must-Have	Core business transaction system; without book check-in and check-out tracking, inventory tracking breaks down completely.
LIB-REQ-04	Automated Overdue Tracking & Fine Calculation Engine	Circulation & Billing	High	Medium	Must-Have	Critical revenue enforcement mechanism and inventory recovery control to prevent material hoarding and long-term asset losses.
LIB-REQ-05	Automated Due Date & Overdue Notifications via Email and SMS	Communications	High	Medium	Should-Have	Drastically minimizes delinquent returns and boosts patron satisfaction; manual reminders and receipt dates serve as MVP workaround.
LIB-REQ-06	Self-Service Book Reservations and Priority Waitlist Queue	Circulation	Medi...	Medium	Should-Have	High demand convenience feature optimizing circulation of high-interest titles; manual staff holds provide an acceptable stopgap.
LIB-REQ-07	Physical Media Barcode & QR Code Optical Scanner Integration	Hardware & Logistics	Medi...	High	Should-Have	Accelerates counter processing throughput and eliminates manual keyboard entry mistakes during peak library operational hours.
LIB-REQ-08	Digital E-Book & Audiobook Integration (OverDrive / Libby API)	Digital Media	Medi...	Low	Could-Have	Extends library reach to remote patrons, but introduces complex DRM licensing agreements and vendor integrations non-essential for launch.
LIB-REQ-09	Personalized AI Reading Recommendations Based on Borrowing History	Patron Experience	Low	Low	Could-Have	Drives patron engagement and discovery, but library functions successfully with curated staff picks and subject searches.
LIB-REQ-10	RFID Automated Smart Shelf Tracking & Real-Time Item Geolocation	Smart Facilities	Low	Low	Won't-Have	Demands prohibitive hardware CAPEX and complex shelf sensory infrastructure with marginal initial ROI for the current lifecycle.

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Result

Thus, the flowchart for obtaining the lexicographically smallest string using character swapping was created successfully using Raptor.